Forecasting How Busy a Venue Will Be, and Counting How Many People Are There

Jayden Bansal · Flock Social LLC, Pennsylvania · Research summary, October 2026

Full paper (29 pages, every data source credited): flockcorp.com/research · KIT Library Seatfinder counts used with permission

Research questions

1. Can a system that watches each venue's own recent live readings beat the popular-times chart at saying how busy one venue will be at one hour? **2.** Can a model small enough to run on a credit-card-sized computer count the people in a room from a 160 × 120 thermal camera, so that busyness can be checked against real people without recording anyone?

What I built

- **The forecast in Flock,** my app for planning nights out. A data pipeline collects hourly live busyness readings for about 1,750 venues in Philadelphia, the Lehigh Valley and Miami. The served number corrects each venue's typical-week curve with a trailing offset from its recent readings and uses the latest reading directly (page 3, Equations 1 and 2). A gradient-boosted model with 106 inputs (every one listed on page 3) is trained and gated against production, and ships only if it wins.
- **Owl,** a 718,550-parameter neural network that counts people on a FLIR Lepton 3.5 thermal camera in 14 to 16 ms on one Raspberry Pi 5 core, trained on a physics-based generator of thermal scenes and six licensed real datasets. It runs inside **Flux**, a venue sensor I designed, where only numbers leave the device, never an image (Figure 1).

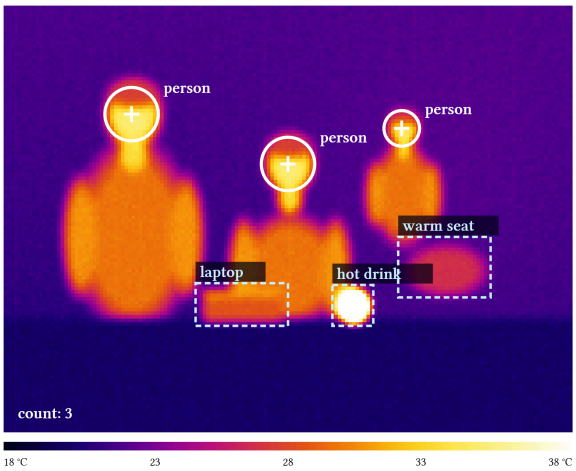


Figure 1. What Owl sees and what it reports. A 160 × 120 thermal frame in degrees Celsius, drawn the way Owl's training generator draws a scene, at the camera's true resolution. Owl puts a point on each person's head and a named box on every other warm thing, here a laptop, a hot drink and a seat someone just left, which a simple heat threshold would count as people. Only the points are counted: this frame counts as three.

How it was tested

Every forecast weight was fitted on earlier days and scored on **119,939 live readings** from 9 September to 8 October 2026 that it never saw, at 1,754 venues, with date-block bootstrap intervals (2,000 resamples of whole dates), persistence baselines and accuracy at every horizon from one to 24 hours. The forecast target is the provider's live reading, a relative 0 to 100 score, so the findings were then checked against real people: **5.7 million hourly counts from 573 counting sites in nine countries**. The core 5.3 million counts (315 sites: pedestrian counters, subway entries, a gym and a university library) were analysed under a plan whose pass rules were fixed and hashed before any data were downloaded; the rest add people counted inside rooms and libraries in Singapore, Barcelona and Karlsruhe. Owl was scored on six real benchmark sets and on sealed sets no version had seen.

Key findings

Result	Flock	Baseline
Exact crowd level (of five), 30-day test, one hour ahead	54.9%	30.2% popular times
Within one crowd level, same readings	78.4%	64.8% popular times
Mean absolute error, same readings	18.00	26.40 (31.8% lower for Flock)
Exact level when the venue was read an hour earlier	77.0%	
Exact level two to 24 hours ahead	33.2–33.6%	30.2%
Real counts: one-hour error, curve + latest departure (15 datasets)	6.41	9.58 curve alone
People inside 153 campus spaces (Wi-Fi users): hour-old count wins at	152 of 153	typical-week curve

Result	Flock	Baseline
People counted in 5 rooms (ROBOD): one-hour error, hour-old count	12.18	21.63 curve
Owl, frames counted exactly, five of six benchmark sets	82–96%	9–77% heat rule
Owl, sealed set of 300 frames no version had seen	94.3%	40.7% heat rule

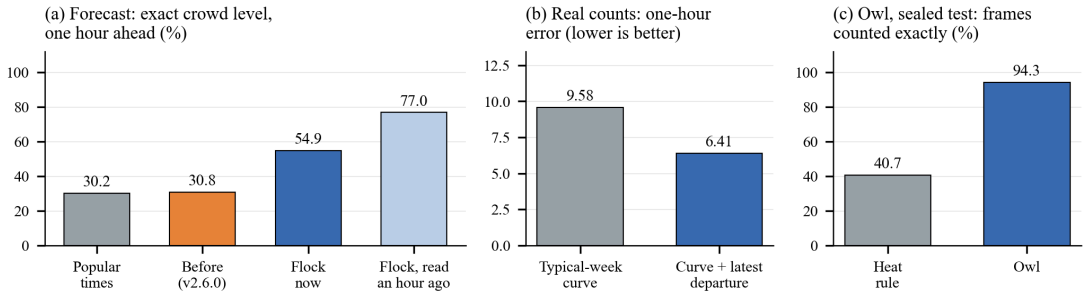


Figure 2. Headline results. (a) Exact crowd level of five, one hour ahead, on 119,939 held-out live readings; "Before" is the model Flock served until 26 September. (b) Mean absolute error, one hour ahead, on the 15 pre-registered public real-count datasets (0 to 100 scale per site); carrying the latest departure from the curve onto the next hour was best in 14 of 15. (c) Owl against the heat-threshold rule on 300 sealed frames.

- **The freshest reading of a venue is worth more than any correction to its typical-week curve.** Where the venue was read in the hour before, those readings carry 92% of the exact-level gain over popular times and 96% of the error reduction, and production now serves that reading directly. Where no fresh reading exists, the trailing offset is the correction.
- **The finding holds for real people.** Wherever the counts are people inside a place (library seats in Karlsruhe, 153 Wi-Fi-covered campus spaces in Barcelona, five rooms counted in Singapore), the hour-old count beats the typical-week curve, as it does on the provider's readings. On street and station counts the typical-week curve is close to calibrated (slope 0.92) and a departure from it lasts for hours, so carrying the latest departure onto the curve (Equation 4) cut one-hour error by 33%. The provider's own curve tracks real movement too: beside 695 New York venues its weekly rhythm matches the nearest subway station's entries (median r 0.74).
- **The popular-times curve alone is a poorly calibrated forecast.** Against live readings, each point of curve is worth about half a point of observed busyness (slope 0.50, r 0.36): quiet hours run 11.5 points busier than the chart says and busy hours 13.9 points quieter.
- **Owl generalises across the cameras it was trained on and, in its stress tests, ignores the warm objects that fool a threshold** (hands, mugs, screens); on crowded classrooms it misses by one person on average, where its predecessor missed by almost five. In an unseen lab room of three to five people (BU-TIV, used with its authors' permission) it counted 34.6% of frames exactly and 79.3% within one person, where the heat rule managed 3.6% and 18.9%. It transfers unevenly to cameras unlike any it trained on: on two low-resolution ceiling sensors and in an atrium of lone, distant people the rule did better.

How Owl counts

Owl is a CenterNet-style network: it predicts a person map $\hat{P}$ over the frame and nine maps for warm objects (hand, pet, hot drink, food, laptop, screen, heater, lamp, warm seat). The count is the number of peaks in the person map above a confidence threshold τ, a peak being a pixel p no lower than any pixel in its neighbourhood $\mathcal{N}(p)$,

$$\hat{N} = \left| \left\{ p : \hat{P}(p) \geq \tau,\ \hat{P}(p) = \max_{q \in \mathcal{N}(p)} \hat{P}(q) \right\} \right| \tag{5}$$

each peak checked against the physics of skin temperature before it counts. It is trained with a penalty-reduced focal loss on the peak maps plus box and density losses, $L = L_{\text{focal}} + \lambda_{\text{box}} L_{\text{box}} + \lambda_{\text{den}} L_{\text{den}}$, using AdamW with a one-cycle schedule, on 40% generated frames and 60% licensed real frames (38,648 frames, 70,402 labelled people). Each checkpoint is chosen on held-out frames, never on training loss.

Scope

- Within Flock, the forecast is scored against the provider's relative 0 to 100 readings, not headcounts; the real-count datasets above are what tie its findings to people. The large gains are one hour ahead.
- Owl's 300-frame sealed set holds one or two people per frame, and its benchmark sets also guided its development.

How the forecast is made

For venue v at hour h on weekday d: $b(v,d,h)$ is the venue's weekly curve; o_v is the median of (observed − curve) over its newest 20 live readings taken before h (within 28 days, at least two); r_v is its latest earlier reading and a that reading's age in hours.

$$\hat{y}_0 = b\left(v, d, h\right) + 0.85 \cdot o_v \tag{1}$$

$$\hat{y} = \left(1 - w_a\right) \cdot \hat{y}_0 + w_a \cdot r_v, \quad w_1 = 1, \; w_a = 0 \quad \text{for} \; a \geq 2 \tag{2}$$

$$\hat{y} = \text{round}\left(\text{clip}\left(b + 0.5\, o_v + \text{clamp}\left(\hat{\delta}, \pm 50\right), 0, 100\right)\right), \quad \hat{\delta} = f\left(x\right), \; x \in \mathbb{R}^{106} \tag{3}$$

$$\hat{y}_{t+1} = b_{t+1} + \left(y_t - b_t\right) \tag{4}$$

Production serves (1) and (2), with weights fitted by mean absolute error on 1 to 5 September and fixed through the 30-day test (then w_2 = 0.85, w_3 = 0.25; both 0 since 13 October). (3) is the delta model, which adds half the offset and its predicted departure $\hat{\delta}$ to the curve: XGBoost, 800 trees at depth 8, learning rate 0.01, with the venue's own curve refused as an input. (4), the latest departure carried onto the next hour's curve, was the best one-hour forecast on 14 of 15 real-count datasets.

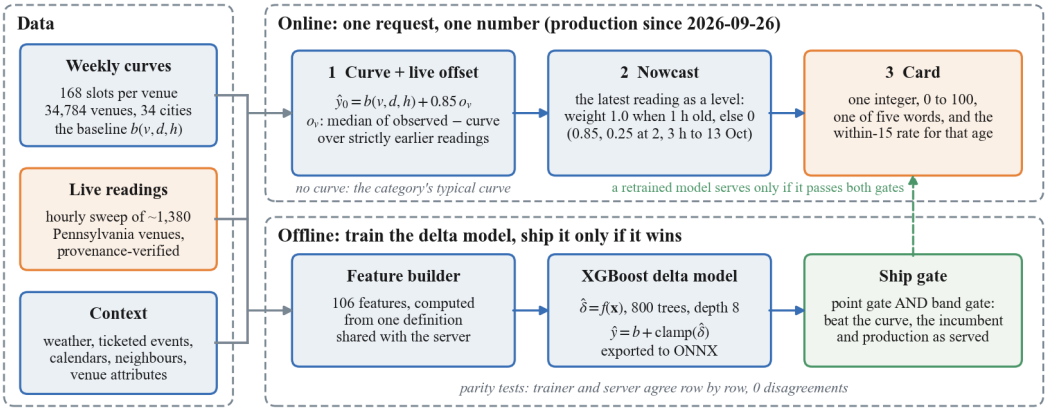


Figure 3. The system. The online path serves every request today. The offline delta model reads 106 inputs and can replace the online path only by passing both halves of the ship gate.

Every input the delta model reads (106)

Time (19)	hour, weekday, month and the sine and cosine of each; weekend; Friday or Saturday after 18:00; lunch, dinner, late night, morning; four season flags
Daylight (3)	daylight hours, hours after sunset, after-sunset flag (solar estimate from latitude and month)
Holidays (6)	holiday, school break, holiday eve, special night, special-night boost and suppress weights (31 national calendars, 1,351 dated special nights)
Weather (18)	temperature, humidity, wind, rain flag; nine condition groups; temperature anomaly against a climate table; four weather interactions
Events (16)	any ticketed event within 2 km; nearest event's attendance, its log and distance; its type as five flags; count, total attendance and its log within 2 km; large event; event × weekend, dinner, bar
Category curves (2)	typical value of the venue's category at this weekday and hour; the same refined by price and rating tier
Neighbours (2)	log count of venues within about 830 m; their mean curve value at this weekday and hour
Venue (5)	category (13 classes), price level, rating, review count and its log
Place types (30)	one flag for each of the 30 most common place types
Feedback (4)	mean crowd report, log report count, has-report flag, mean error of past predictions (verified in-app reports, 28 days)
Row type (1)	live or typical-week row
Added for the retrain	*live readings (5): the latest earlier reading's departure from the curve and its age, the trailing offset and its count, the curve one hour earlier; sports games (6): a tracked team plays today, number of games, evening game, home game, distance to the arena, arena within 10 km*

What comes next

A pre-registered retrain in early November 2026, scored on 14 held-out days against thresholds fixed in code on 25 and 26 September, before its data existed (the production number it must beat was simplified on 13 October, which makes the test harder); a sealed test on Flux's own camera in rooms of one to fifteen people; and a venue study that pairs hand counts with Owl and the provider's live scores, to measure how busyness scores track the people inside.